

Notes: Heckman & Vytlačil (Econometrica, 2005)

Structural Equations, Treatment Effects, and Econometric Policy Evaluation

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1 Big picture

- **Question.** How should economists evaluate policies when treatment responses are heterogeneous, selection into treatment is endogenous, and different estimators identify different margins?
- **Core contribution.** The paper uses the **marginal treatment effect** (MTE) to connect two traditions: structural econometrics and the treatment-effects literature.
- **Main idea.** Instead of treating ATE, treatment-on-treated, LATE, IV, OLS, and policy-relevant treatment effects as unrelated objects, the paper shows that many of them are weighted averages of the same underlying object, the MTE.
- **Key object.** The MTE is the mean gain from treatment for people at a particular margin of selection into treatment:

$$\text{MTE}(x, u_D) = E(Y_1 - Y_0 \mid X = x, U_D = u_D).$$

- **Why it matters.** If gains vary across people and people select into treatment partly based on those gains, there is no single treatment effect. The parameter depends on which people the estimator emphasizes.
- **Central warning about IV.** In heterogeneous-response models, a valid instrument does not simply identify “the treatment effect.” It identifies a weighted average of MTEs, and the weights depend on the instrument.
- **Policy implication.** A policy evaluator must ask: Which margin changes under the policy? A conventional IV estimate may not answer that question unless its weights match the policy-relevant weights.
- **Structural-econometric implication.** The paper gives a nonparametric version of the structural concept of policy invariance: the MTE can be stable across policy environments if the underlying outcome equations and choice unobservables are stable.
- **Limitation.** The approach requires an ordered scalar index in the selection equation. When both treatment responses and choices are heterogeneous in fully general ways, IV methods lose clear economic interpretation and stronger structural assumptions are needed.

2 Data, objects, and measurement in the theory

2.1 This is a conceptual and econometric paper

The paper is not an empirical application with a new dataset. It is a theoretical econometrics paper about how to define, identify, and interpret policy-evaluation parameters.

- The “data” in the analysis are generic observed variables:

$$(Y, D, X, Z),$$

where Y is the observed outcome, D is treatment status, X are observed covariates, and Z are variables shifting treatment choice.

- The unobserved objects are potential outcomes Y_1, Y_0 and latent variables driving treatment selection.
- The examples and figures use a generalized Roy model calibrated to generate transparent differences between MTE, IV, OLS, and policy-relevant parameters.

Interpretation: The paper’s goal is not to estimate one empirical effect. Its goal is to define the correct economic object and show how common estimators map into that object.

2.2 Potential outcomes and observed outcome

The standard potential-outcome notation is

$$Y_1 \text{ if treated, } \quad Y_0 \text{ if untreated.}$$

The observed outcome is

$$Y = DY_1 + (1 - D)Y_0. \quad (1)$$

Interpretation: We observe only one potential outcome for each person. Policy evaluation is about learning the missing counterfactual and aggregating treatment gains for the policy-relevant group.

2.3 Treatment selection

The paper embeds potential outcomes inside a latent-index choice equation. The generic choice rule is

$$D = 1[\mu_D(Z) - V \geq 0], \quad (2)$$

where V is an unobserved resistance or cost of participation.

Define the propensity score

$$P(Z) = \Pr(D = 1 \mid Z). \quad (3)$$

After a monotone normalization, the selection rule can be written as

$$D = 1[P(Z) \geq U_D], \quad U_D \sim \text{Uniform}(0, 1). \quad (4)$$

Interpretation: The scalar U_D ranks individuals by their resistance to treatment. People with low U_D are more likely to participate; people with high U_D require stronger incentives or instruments to participate.

2.4 The policy margin

A marginal policy change shifts the probability of treatment. The people affected are those near the new margin. The MTE evaluates gains at each value of that margin:

$$\text{MTE}(x, u_D) = E(Y_1 - Y_0 \mid X = x, U_D = u_D). \quad (5)$$

Interpretation: If people with low resistance select into treatment because they have high gains, then $\text{MTE}(x, u_D)$ may decline in u_D . If selection is adverse, it could rise. The shape of the MTE is therefore economically meaningful.

2.5 Support and identification

Identification of the full MTE curve requires variation in $P(Z)$ over the relevant range of u_D .

- If the instrument changes the propensity score only in a narrow region, only a local part of the MTE is identified.
- ATE, TT, TUT, and broad policy counterfactuals require sufficient support over the margins they weight.
- If support is incomplete, the MTE approach clarifies what is identified and what requires extrapolation.

Interpretation: The paper's framework separates two questions often conflated in applied work: what object the estimator targets and whether the available variation identifies enough of that object.

3 Models and empirical specifications

3.1 Benchmark structural model

A prototypical structural treatment model writes potential outcomes as

$$Y_1 = \mu_1(X, U_1), \quad (6)$$

$$Y_0 = \mu_0(X, U_0), \quad (7)$$

$$D = 1[\mu_D(Z) - V \geq 0]. \quad (8)$$

Interpretation: This is a structural system because it specifies both outcome equations and the choice equation. The structural approach can answer counterfactual policy questions if its invariance and exogeneity assumptions are credible.

3.2 The MTE as the primitive treatment parameter

The marginal treatment effect is

$$\text{MTE}(x, u_D) = E(Y_1 - Y_0 \mid X = x, U_D = u_D). \quad (9)$$

Interpretation: The MTE is a local willingness-to-pay or marginal-gain curve. It tells us the expected gain for people who would be indifferent to treatment at a given propensity-score margin.

3.3 Average treatment effect

The average treatment effect conditional on $X = x$ is

$$\text{ATE}(x) = \int_0^1 \text{MTE}(x, u_D) du_D. \quad (10)$$

Interpretation: ATE gives equal weight to all latent-resistance types. It is the effect of assigning treatment randomly across the full population at covariate value x .

3.4 Treatment on the treated and untreated

Treatment on the treated is

$$\text{TT}(x) = \int_0^1 \text{MTE}(x, u_D) h_{TT}(x, u_D) du_D, \quad (11)$$

where the weights emphasize persons who actually select into treatment.

Treatment on the untreated is

$$\text{TUT}(x) = \int_0^1 \text{MTE}(x, u_D) h_{TUT}(x, u_D) du_D. \quad (12)$$

Interpretation: TT and TUT differ when selection is related to gains. TT is the gain for those induced or choosing treatment under the status quo; TUT is the gain for those not treated under the status quo.

3.5 Local average treatment effect

For two margins u_D and u'_D , the local average treatment effect is

$$\text{LATE}(x; u_D, u'_D) = \frac{1}{u_D - u'_D} \int_{u'_D}^{u_D} \text{MTE}(x, u) du. \quad (13)$$

Interpretation: LATE averages MTE only over compliers moved by a particular instrument shift. Different instruments move different margins and therefore identify different LATEs.

3.6 Local instrumental variables

The paper's local-IV result links the derivative of the conditional mean outcome with respect to the propensity score to the MTE:

$$\frac{\partial E(Y \mid X = x, P(Z) = p)}{\partial p} = \text{MTE}(x, p). \quad (14)$$

Interpretation: This is the central identification result. If the propensity score varies continuously, the slope of the outcome-propensity relationship recovers the MTE at that margin.

3.7 Policy relevant treatment effect

For a policy that changes the distribution of the propensity score from P to P^* , the policy relevant treatment effect can be written as

$$\text{PRTE}(x) = \int_0^1 \text{MTE}(x, u_D) h_{\text{PRTE}}(x, u_D) du_D. \quad (15)$$

Interpretation: PRTE is the right object when the policy does not randomly assign treatment but changes the incentives or constraints determining participation. The weights come from the policy-induced change in the participation margin.

3.8 IV as a weighted average of the MTE

For an instrument function $J(Z)$, the linear IV estimand conditional on $X = x$ can be expressed as

$$\beta_{IV}(x; J) = \int_0^1 \text{MTE}(x, u_D) h_{IV}(u_D \mid x; J) du_D. \quad (16)$$

The weights are determined by how $J(Z)$ covaries with the treatment propensity and with the event that the latent resistance is below the propensity score.

Interpretation: The same dataset and same valid exclusion restrictions can deliver different IV estimands when the instrument changes. This is not a nuisance. It is the economic content of IV under heterogeneity.

3.9 OLS as a weighted average plus selection bias

The paper also writes OLS as a weighted average of MTEs with weights affected by selection into treatment and untreated outcomes.

Interpretation: OLS need not be a simple average treatment effect, but IV is not automatically closer to the policy-relevant object. In heterogeneous models, the relevant comparison is not “OLS biased, IV unbiased” but “which estimand answers which policy question?”

3.10 The fundamental asymmetry of IV

The paper emphasizes that IV methods can allow heterogeneous responses while retaining a scalar monotone choice equation. But if the choice equation is also heterogeneous in a fully general way, the IV approach breaks down.

Interpretation: The reason is economic: IV needs to order people along a single participation margin. If different instruments move different people in nonordered ways, there is no single MTE curve to integrate over.

4 Identification and empirical design

4.1 Assumptions behind the MTE framework

The paper’s nonparametric selection model relies on five broad ingredients:

- potential outcomes exist and are separable from the realized treatment indicator;
- treatment choice follows a scalar latent-index rule;
- instruments shift participation but are exogenous to potential outcomes conditional on covariates;
- the propensity score has enough support to trace the relevant margins;
- policy invariance holds for the structural outcome and choice components used in counterfactual prediction.

Interpretation: The assumptions are weaker than fully parametric structural models but stronger than design-based treatment evaluation for one local effect.

4.2 What makes the approach policy-relevant

- The MTE is tied to the choice margin, not just to an arbitrary comparison of treated and untreated groups.
- Policy changes typically shift incentives or eligibility, so they affect the distribution of $P(Z)$.
- Once the MTE is identified, a policy evaluator can recompute weights for a new policy and obtain the policy-relevant effect.

Interpretation: The framework shows why a parameter estimated from one instrument does not necessarily forecast another policy. The relevant parameter changes when the affected margin changes.

4.3 Internal validity versus external validity

The treatment-effects literature often focuses on internal validity: estimating a treatment parameter in an existing environment without selection bias. The structural approach emphasizes external validity: forecasting new policies and new environments. The MTE framework sits between them.

Interpretation: The paper’s point is not that internal validity is unimportant. It is that internal validity for the wrong margin does not answer a policy question about a different margin.

4.4 Testable implications

The latent-index model imposes restrictions, including:

- conditional mean outcomes should depend on instruments through the propensity score rather than through Z directly;
- monotonicity and index restrictions should hold for the participation equation;
- curvature of $E(Y \mid P(Z) = p)$ is informative about heterogeneity in the MTE.

Interpretation: Unlike a purely agnostic treatment-effects setup, the MTE framework has empirical content: it restricts how outcomes and participation can vary with instruments.

4.5 What the paper cannot solve without stronger structure

- If the choice equation has multiple unordered dimensions of unobserved heterogeneity, a scalar MTE is insufficient.
- If the policy changes equilibrium prices, peer effects, or general equilibrium feedback, additional modeling is needed.
- If the support of $P(Z)$ is narrow, the full ATE or broad PRTE requires extrapolation.
- If instruments change preferences, information, or outcomes directly, MTE interpretation fails.

Interpretation: The paper clarifies the boundary between semiparametric policy evaluation and fully structural modeling.

5 Tables and figures

5.1 Tables IA and IB: treatment parameters as weighted averages of the MTE

Read:

- Table IA writes ATE, TT, LATE, TUT, PRTE, IV, and OLS as weighted averages of $\text{MTE}(x, u_D)$.
- Table IB gives the corresponding weights.
- ATE uses uniform weights over u_D .
- TT places more weight on people who are more likely to be treated under the status quo.
- TUT places more weight on people who are less likely to be treated.
- PRTE weights are determined by how the policy changes the distribution of the propensity score.
- IV weights depend on the chosen instrument, not just on the treatment itself.

Economic message: These two tables are the core map of the paper. They show that familiar estimands are not competing definitions floating independently; they are different aggregations of the same marginal-gain curve.

TABLE IA
TREATMENT EFFECTS AND ESTIMANDS AS WEIGHTED
AVERAGES OF THE MARGINAL TREATMENT EFFECT

$$\begin{aligned} \text{ATE}(x) &= \int_0^1 \Delta^{\text{MTE}}(x, u_D) du_D \\ \text{TT}(x) &= \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{\text{TT}}(x, u_D) du_D \\ \text{LATE}(x, u_D, u_D) &= \frac{1}{u_D - u_D^0} \left[\int_{u_D^0}^{u_D} \Delta^{\text{MTE}}(x, u) du \right] \\ \text{TUT}(x) &= \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{\text{TUT}}(x, u_D) du_D \\ \text{PRTE}(x) &= \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{\text{PRTE}}(x, u_D) du_D \\ \text{IV}(x) &= \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{\text{IV}}(x, u_D) du_D \\ \text{OLS}(x) &= \int_0^1 \Delta^{\text{MTE}}(x, u_D) h_{\text{OLS}}(x, u_D) du_D \end{aligned}$$

(a) Table IA: estimands as MTE averages.

TABLE IB
WEIGHTS

$$\begin{aligned} h_{\text{ATE}}(x, u_D) &= 1 \\ h_{\text{TT}}(x, u_D) &= \left[\int_{u_D}^1 f(p|X=x) dp \right] \frac{1}{E(P|X=x)} \\ h_{\text{TUT}}(x, u_D) &= \left[\int_0^{u_D} f(p|X=x) dp \right] \frac{1}{E((1-P)|X=x)} \\ h_{\text{PRTE}}(x, u_D) &= \left[\frac{F_{P^*,X}(u_D|x) - F_{P,X}(u_D|x)}{\Delta \hat{P}(x)} \right], \text{ where } \Delta \hat{P}(x) = E(P|X=x) - E(P^*|X=x) \\ h_{\text{IV}}(x, u_D) &= \left[\int_{u_D}^1 (p - E(P|X=x)) f(p|X=x) dp \right] \frac{1}{\text{Var}(P|X=x)} \text{ for } P(Z) \text{ as an instrument} \\ h_{\text{OLS}}(x, u_D) &= 1 + \frac{E(U_1|X=x, U_D=u_D)h_1(x, u_D) - E(U_0|X=x, U_D=u_D)h_0(x, u_D)}{\Delta^{\text{MTE}}(x, u_D)}, \\ &\quad \text{if } \Delta^{\text{MTE}}(x, u_D) \neq 0, \\ &\quad = 0 \text{ otherwise} \\ h_1(x, u_D) &= \left[\int_{u_D}^1 f(p|X=x) dp \right] \left[\frac{1}{E(P|X=x)} \right] \\ h_0(x, u_D) &= \left[\int_0^{u_D} f(p|X=x) dp \right] \frac{1}{E((1-P)|X=x)} \end{aligned}$$

(b) Table IB: weights applied to the MTE.

Additional interpretation. Table IA is useful because it forces the reader to ask which population or margin is being averaged. Table IB is useful because it makes the same point operational: two estimators can both be valid and still answer different questions because their weights differ. The tables also show why policy evaluation is a weighting problem once the MTE is known.

5.2 Figure 1A, Figure 1B, and Table II: MTE, IV, OLS, and Roy-model examples

Read:

- Figure 1A plots weights for several treatment parameters in a generalized Roy model.
- Figure 1B plots the declining MTE together with linear IV and OLS weights.
- Table II reports the numerical treatment parameters and estimands generated by the example.
- Treatment on the treated exceeds the ATE, while treatment on the untreated is lower, reflecting positive selection on gains.
- OLS differs from TT because it also incorporates selection in baseline outcomes.
- Linear IV does not automatically recover ATE, TT, or PRTE; it recovers the MTE weighted by the instrument-induced margin.

Economic message: The Roy example makes the abstract weighting formulas concrete. When high-gain individuals are more likely to select into treatment, the treated and untreated populations have different gains, and an IV estimate depends on which margin the instrument shifts.

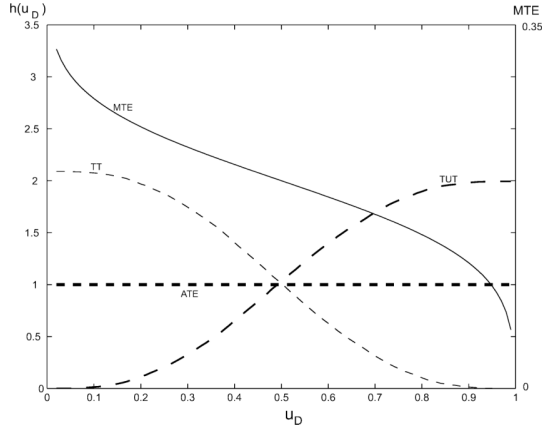


FIGURE 1A.—Weights for the marginal treatment effect for different parameters.

(a) Figure 1A: weights for treatment parameters.

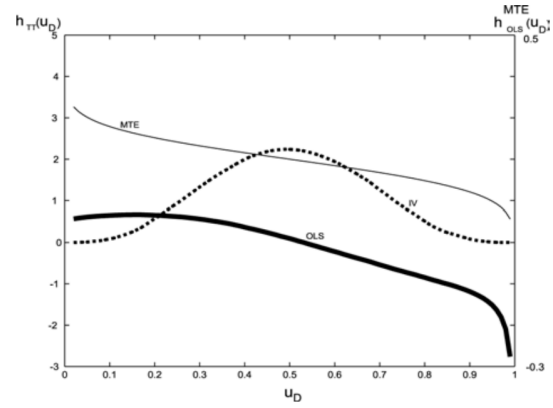


FIGURE 1B.—Marginal treatment effect vs. linear instrumental variables and ordinary least squares weights. Model used to generate Figures 1A and 1B:

(b) Figure 1B: MTE, IV, and OLS weights.

TABLE II
TREATMENT PARAMETERS AND ESTIMANDS IN THE
GENERALIZED ROY EXAMPLE

Treatment on the treated	0.2353
Treatment on the untreated	0.1574
Average treatment effect	0.2000
Sorting gain ^a	0.0353
Policy relevant treatment effect (PRTE)	0.1549
Selection bias ^b	-0.0628
Linear instrumental variables ^c	0.2013
Ordinary least squares	0.1725

^a $TT - ATE = E(Y_1 - Y_0 | D = 1) - E(Y_1 - Y_0)$.

^b $OLS - TT = E(Y_0 | D = 1) - E(Y_0 | D = 0)$.

^c Using propensity score $P(Z)$ as the instrument.

Note: The model used to create Table II is the same as those used to create Figures 1A and 1B. The PRTE is computed using a policy t characterized as follows:
If $Z > 0$ then $D = 1$ if $Z(1 + t) - V > 0$.
If $Z \leq 0$ then $D = 1$ if $Z - V > 0$.
For this example t is set equal to .2.

Table II: treatment parameters and estimands in the generalized Roy example.

Additional interpretation. Figure 1B is especially important because it shows visually why “the IV estimate” is not a universal causal effect. The IV weight is concentrated where the instrument changes participation. If the MTE is sloped, moving the weight changes the estimand. Table II then quantifies the problem: ATE, TT, TUT, OLS, IV, and PRTE are close enough to be comparable but not identical enough to be interchangeable.

5.3 Figures 2A and 2B: local instrumental variables and the derivative of the conditional mean

Read:

- Figure 2A plots $E(Y | P(Z) = p)$.
- If the MTE does not vary across u_D , the conditional mean is approximately linear in p .
- If MTE varies across the selection margin, $E(Y | P(Z) = p)$ is curved.
- Figure 2B plots the derivative of Figure 2A; that derivative is the identified MTE.

Economic message: These figures show the intuition behind local IV. The MTE is not estimated by a level difference; it is recovered from the slope of the outcome-propensity relationship.

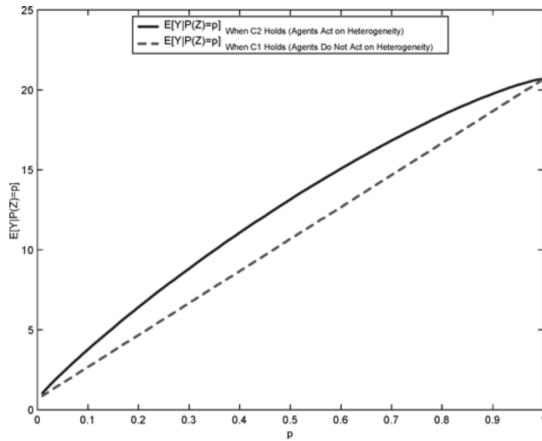


FIGURE 2A.—Plot of the $E(Y|P(Z) = p)$.

(a) Figure 2A: $E(Y | P(Z) = p)$.

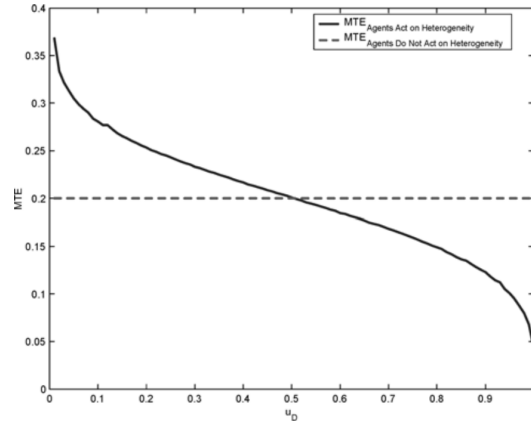


FIGURE 2B.—Plot of the identified marginal treatment effect from Figure 2A (the derivative). :: Parameters for the general heterogeneous case are the same as those used in Figures 1B and 1C. For the homogeneous case we impose $U_1 = U_0$ ($\sigma_1 = \sigma_0 = 0.012$).

(b) Figure 2B: derivative identifies MTE.

Additional interpretation. The figures explain why flexible estimation of the conditional mean can be informative about heterogeneity. If the conditional mean is linear, a constant treatment effect is plausible. If it is curved, marginal returns vary across the participation margin.

5.4 Figure 3A and Table III: IV weights versus policy-relevant weights

Read:

- Figure 3A compares the MTE with weights for OLS, linear IV, and a policy-relevant treatment effect.
- The policy-relevant weight is tied to a specified policy shift, not to an arbitrary instrument.
- Table III reports IV estimands using different instruments and compares them to the policy relevant treatment effect.
- The estimates differ because the instruments put weight on different parts of the MTE curve.

Economic message: The visual and table jointly show that instrument choice is policy choice in disguise. An instrument is useful only insofar as its induced margin corresponds to the policy margin one cares about.

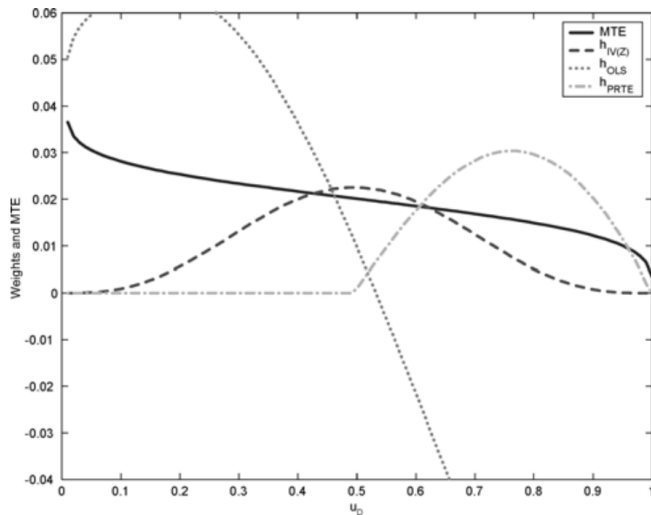


FIGURE 3A.—Marginal treatment effect vs. linear instrumental variables, ordinary least squares, and policy relevant treatment effect weights when $P(Z)$ is the instrument for the policy variable at the base of Table II.

(a) Figure 3A: MTE versus OLS, IV, and PRTE weights.

TABLE III
LINEAR INSTRUMENTAL VARIABLE ESTIMANDS AND THE POLICY
RELEVANT TREATMENT EFFECT

Using propensity score $P(Z)$ as the instrument	0.2013
Using propensity score $P(Z(1 + t(1[Z > 0])))$ as the instrument	0.1859
Using a dummy B as an instrument ^a	0.1549
Policy relevant treatment effect (PRTE)	0.1549

^aThe dummy B is such that $B = 1$ if an individual belongs to a randomly assigned eligible population and 0 otherwise.

(b) Table III: IV estimands and PRTE.

Additional interpretation. Table III is a compact warning against reporting an IV coefficient without interpreting the margin. Even when every instrument is valid, each instrument asks a different question. Figure 3A shows the weighting mismatch that drives the numerical differences.

5.5 Figures 3B and 3C: policy instruments and ideal randomized eligibility

Read:

- Figure 3B compares IV weights from different transformations of the policy variable.
- Figure 3C compares the policy-relevant weights with weights from a randomized eligibility instrument.
- An ideal randomized policy instrument can match the PRTE margin more closely than a generic IV.
- The figures emphasize that the identity of compliers depends on the policy design.

Economic message: Randomization is not automatically the answer to every policy question. It is powerful when the randomization moves the same population margin that the target policy would move.

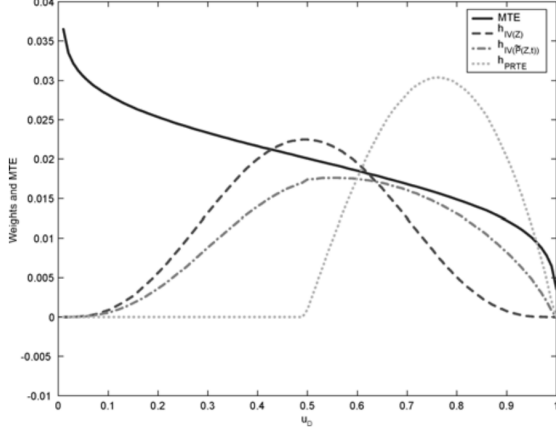


FIGURE 3B.—Marginal treatment effect vs. linear IV with Z as an instrument, linear IV with $(1 + t(\mathbb{1}[Z > 0])) = \hat{P}(Z, t)$ as an instrument, and policy relevant treatment effect weights for the policy defined at the base of Table II.

(a) Figure 3B: IV weights under alternative policy instruments.

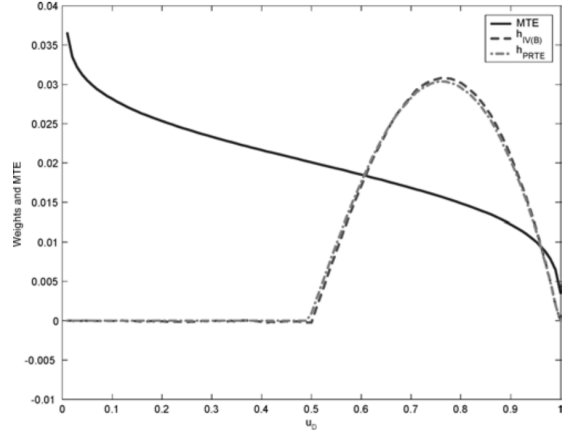


FIGURE 3C.—Marginal treatment effect vs. IV policy and policy relevant treatment effects for the policy defined at the base of Table II.

(b) Figure 3C: ideal IV and PRTE weights.

Additional interpretation. Figures 3B and 3C clarify the difference between a design and a target. A well-designed experiment can recover a policy-relevant object, but only if the experiment’s eligibility or incentive variation reproduces the policy-induced selection margin.

5.6 Figure 4 and Table IV: negative IV weights and monotonicity

Read:

- Figure 4 shows that IV weights can be negative under some instrument distributions.
- Negative weights arise when monotonicity does not hold globally or when the instrument shifts participation in conflicting directions across groups.
- Table IV reports how the IV estimand changes as the covariance structure changes.
- Even with positive individual treatment effects, negative weights can make the IV estimand difficult to interpret as an average treatment effect.

Economic message: The monotonicity condition is not a technical footnote. It is what lets IV be interpreted as a weighted average of gains for a coherent complier margin.

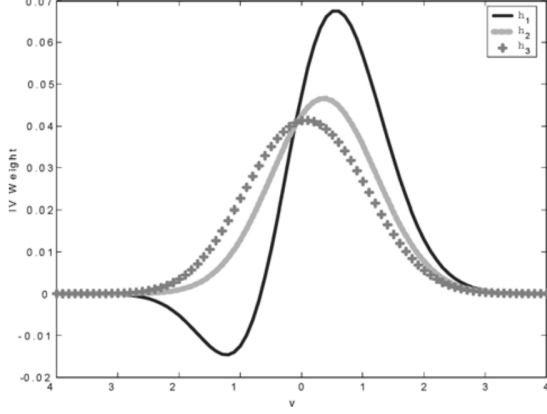


FIGURE 4.—IV weights when $Z \sim p_1 N(\mu_1, \Sigma_1) + p_2 N(\mu_2, \Sigma_2)$ for different values of Σ_2 used to generate Figure 4:

(a) Figure 4: negative IV weights.

TABLE IV
THE IV ESTIMATOR AND $\text{Cov}(Z_1, \beta' Z)$ ASSOCIATED WITH EACH VALUE OF Σ_2
(GROUP 2 COVARIANCE)

Weights	Σ_2	IV	$\text{Cov}(Z_1, \beta' Z) = \beta' \Sigma_2^1$
h_1	$\begin{bmatrix} 0.6 & -0.3 \\ -0.3 & 0.6 \end{bmatrix}$	0.133	-0.30
h_2	$\begin{bmatrix} 0.6 & -0.1 \\ -0.1 & 0.6 \end{bmatrix}$	0.177	-0.02
h_3	$\begin{bmatrix} 0.6 & 0.1 \\ 0.1 & 0.6 \end{bmatrix}$	0.194	0.26

Weights for mixture of normals IV:

$$h_{IV}(v) = \frac{\left[\frac{P_1 \beta' \Sigma_1^1}{(\beta' \Sigma_1 \beta + \sigma_p^2)^{1/2}} \exp\left[-\frac{1}{2} \left(\frac{v - \beta' \mu_1}{(\beta' \Sigma_1 \beta + \sigma_p^2)^{1/2}} \right)^2\right] + \frac{P_2 \beta' \Sigma_2^1}{(\beta' \Sigma_2 \beta + \sigma_p^2)^{1/2}} \exp\left[-\frac{1}{2} \left(\frac{v - \beta' \mu_2}{(\beta' \Sigma_2 \beta + \sigma_p^2)^{1/2}} \right)^2\right] \right] f_V(v)}{\frac{P_1 \beta' \Sigma_1^1}{(\beta' \Sigma_1 \beta + \sigma_p^2)^{1/2}} \exp\left[-\frac{1}{2} \left(\frac{v - \beta' \mu_1}{(\beta' \Sigma_1 \beta + \sigma_p^2)^{1/2}} \right)^2\right] + \frac{P_2 \beta' \Sigma_2^1}{(\beta' \Sigma_2 \beta + \sigma_p^2)^{1/2}} \exp\left[-\frac{1}{2} \left(\frac{v - \beta' \mu_2}{(\beta' \Sigma_2 \beta + \sigma_p^2)^{1/2}} \right)^2\right]}$$

where Σ_1^1 and Σ_2^1 are the first rows of Σ_1 and Σ_2 , respectively. Clearly, $h_{IV}(-\infty) = 0$ and $h_{IV}(\infty) = 0$. The weights clearly integrate to 1 over the support of $V = (-\infty, \infty)$. Observe that if $P_2 = 0$, the weights must be positive. Thus the structure of the covariances of the instruments is a key determinant of the positivity of the weights for any instrument. It has nothing to do with the *ceteris paribus* effect of Z_1 on $P(Z)$ in the general case (changing Z_1 holding all other components of Z fixed). Now observe that a necessary condition for $h_{IV} < 0$ is that $\text{sign}(\beta' \Sigma_1^1) = -\text{sign}(\beta' \Sigma_2^1)$, i.e., that the covariance between Z_1 and $\beta' Z$ be of opposite signs in the two populations. Without loss of generality assume that $\beta' \Sigma_1^1 > 0$. If it equals zero, we fail the rank condition. $f_V(v)$ is the density of V .

(b) Table IV: IV estimands and covariance changes.

Additional interpretation. Figure 4 is one of the paper's strongest cautions. If the instrument does not move everyone in the same direction, IV is not simply a local average over compliers. Table IV shows that changing the covariance structure can change the sign and magnitude of the implicit weights, making the estimand sensitive to distributional details of the instrument.

5.7 Table V: comparing structural, treatment-effect, and MTE-based approaches

Read:

- The structural approach gives interpretable economic parameters and can address many counterfactual policies, but requires stronger assumptions.
- The conventional treatment-effect approach emphasizes credible internal validity for a specific treatment effect, but its link to welfare and extrapolation is often limited.
- The MTE approach gives interpretable weighted averages tied to willingness to pay and policy margins.
- The MTE approach can generate multiple treatment parameters, but only when the required support and invariance assumptions are credible.

Economic message: Table V is the paper's conceptual synthesis. It shows that MTE is meant to keep the interpretability of structural econometrics while retaining the transparency of treatment-effects methods.

TABLE V
COMPARISON OF ALTERNATIVE APPROACHES TO PROGRAM EVALUATION

	Structural Econometric Approach	Treatment Effect Approach	Approach Based on MTE
Interpretability	Well defined economic parameters and welfare comparisons	Link to economics and welfare comparisons obscure	Interpretable in terms of willingness to pay; weighted averages of the MTE answer well-posed economic questions
Range of questions addressed	Answers many counterfactual questions	Focuses on one treatment effect or narrow range of effects	With support conditions, generates all treatment parameters
Extrapolation to new environments	Provides ingredients for extrapolation	Evaluates one program in one environment	Can be partially extrapolated; extrapolates to new policy environments with different distributions of the probability of participation due solely to differences in distributions of Z
Comparability across studies	Policy invariant parameters comparable across studies	Not generally comparable	Partially comparable; comparable across environments with different distributions of the probability of participation due solely to differences in distributions of Z
Key econometric problems	Exogeneity, policy invariance, and selection bias	Selection bias	Selection bias: exogeneity and policy invariance if used for forecasting
Range of policies that can be evaluated	Programs with either partial or universal coverage, depending on variation in data (prices/endowments)	Programs with partial coverage (treatment and control groups)	Programs with partial coverage (treatment and control groups)
Extension to general equilibrium evaluation	Need to link to time series data; parameters compatible with general equilibrium theory	Difficult because link to economics is not precisely specified	Can be linked to non-parametric general equilibrium models under exogeneity and policy invariance

Table V: comparison of approaches to program evaluation.

Additional interpretation. The table frames the paper as a bridge rather than a replacement. Structural models are broad but assumption-heavy. Treatment-effect estimators are credible but narrow. MTE-based evaluation provides a disciplined middle ground: it makes weights explicit, connects estimates to policy margins, and states where extrapolation begins.

6 Short summary

- The paper develops the MTE as a unifying object for policy evaluation.
- The MTE is the average gain from treatment for people at a specified latent participation margin.
- ATE, TT, TUT, LATE, PRTE, IV, OLS, and matching estimands can be written as weighted averages of the MTE.
- Different estimators differ because they use different weights over the latent resistance to treatment.
- Local IV identifies the MTE as the derivative of the conditional mean outcome with respect to the propensity score.
- Linear IV estimates an instrument-specific weighted average of the MTE, not generally ATE or TT.
- Policy-relevant treatment effects require weights generated by the policy change itself.
- Negative weights and failures of monotonicity can make IV estimates hard to interpret as average gains.
- The MTE approach links treatment-effect methods to structural policy evaluation through a nonparametric policy-invariant object.

- The approach requires a scalar index selection model and sufficient support; with general heterogeneity in both choice and outcomes, stronger structural models are needed.

7 Conclusion

7.1 Core conclusion

Heckman and Vytlacil's central conclusion is that policy evaluation should be organized around the margins affected by policy. The MTE is the primitive object that describes treatment gains along the participation margin. Once the MTE is known, many familiar treatment parameters are obtained by applying different weights.

7.2 Why the paper changes how to read IV estimates

The paper changes the interpretation of IV in heterogeneous treatment-effect models. IV is not merely a way to remove selection bias and recover a universal causal effect. Instead, IV defines a margin. The coefficient is a weighted average of MTEs for the people moved by that instrument. Therefore, asking whether an instrument is valid is not enough. One must also ask whether the instrument answers the policy question of interest.

7.3 Economic intuition

Selection into treatment is often purposeful. People participate in schooling, job training, migration, or programs because their perceived gains and costs differ. If gains are heterogeneous, then those at the low-cost margin may not have the same returns as those at the high-cost margin. A policy that subsidizes participation changes which margin participates. The MTE maps each margin to its gain.

7.4 Implication for applied empirical work

Applied researchers should report not just a coefficient but also the margin that the coefficient represents. In an IV design, this means asking who the compliers are and what part of the MTE curve the instrument weights. In a policy evaluation, it means comparing the estimator's weights with the policy-relevant weights. A beautifully identified local estimate may be irrelevant for a policy that changes a different margin.

7.5 Implication for external validity

The paper gives a precise language for external validity. To extrapolate from one environment to another, one needs an invariant MTE and enough information about how the new policy changes participation. Without support over the new margin, extrapolation requires additional assumptions. Thus, the framework does not magically solve external validity, but it shows exactly what must be transported.

7.6 Implication for structural econometrics

The paper also disciplines structural work. It shows that one can obtain economically interpretable policy parameters without fully parametric assumptions, provided a scalar-index selection structure is credible. At the same time, it defends the structural tradition by showing where treatment-effects methods are insufficient: broad counterfactuals, general equilibrium policies, and heterogeneous choice equations often require more structure.

7.7 Policy takeaway

For policy makers, the lesson is that “the effect of a program” is not a single number unless the affected group is specified. Expanding a program, shrinking a program, changing eligibility, changing subsidies, and random assignment all target different people. The MTE framework makes those differences explicit and therefore helps avoid using the wrong treatment effect for the policy at hand.

7.8 Research agenda

The paper points toward several research tasks:

- estimating MTE curves flexibly rather than reporting only single IV coefficients;
- designing experiments so that the randomized margin matches the policy margin;
- studying support conditions before making external-validity claims;
- integrating MTE methods with equilibrium models when policies affect prices, peer behavior, or aggregate constraints;
- developing tools for settings where treatment choices are multidimensional and cannot be ordered by a single latent index.

7.9 Final takeaway

The enduring message is that treatment effects are about margins. The MTE turns this insight into a formal framework. It explains why different estimators give different answers, when those answers are economically meaningful, and how to construct a treatment parameter that is relevant for a specific policy.